

Project Overview

Build a comprehensive task management application where teams can collaborate on projects and tasks with secure access controls.

User Entities

1. User

- username (unique identifier)
- email (contact)
- password (hashed)
- created_date
- last_login
- role (admin, manager, member)

2. Project

- project_name
- description
- owner (User reference)
- created_date
- status (active, archived)
- team_members

3. Task

- title
- description
- assigned_to (User reference)
- project_id (Project reference)
- status (todo, in_progress, completed, blocked)
- priority (low, medium, high)
- due_date
- created_date
- updated_date
- comments

Core Requirements

1. Authentication & Security

- Users login with JWT tokens
- Token-based API authentication
- Role-based access control (RBAC):
 - Admin: Full system access
 - Manager: Project and team management
 - Member: Task assignment and updates

- Password hashing and validation

2, Database & Persistence

- Use PostgreSQL for reliable data persistence
- Support for complex queries and transactions
- Automatic schema generation from entities

3. API & Documentation

- RESTful API with standard CRUD operations
- API documentation with Swagger/OpenAPI
- Clear error messages for all failure scenarios

CRUD Operations

Each entity should support:

- CREATE: Add new projects, tasks, users
- READ: List and retrieve by ID
- UPDATE: Edit project details, task status, user information
- DELETE: Remove completed tasks, archived projects